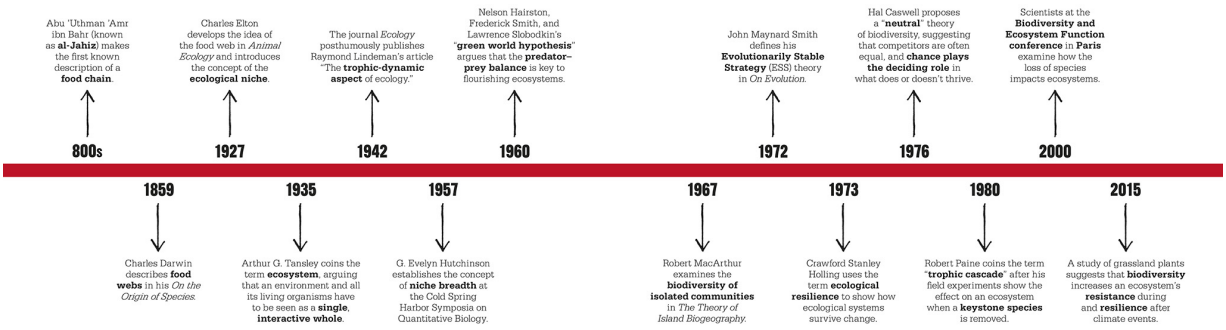
echinoderm was shown to be a keystone species, playing a crucial role within its ecosystem.



Read accessible description

Island isolation

Habitat fragmentation is now a major problem in most terrestrial environments because it leaves specialist organisms isolated. For that reason, research into the biogeography of islands—those surrounded by ocean but also “islands” of distinct habitat surrounded by a very different environment—is so important in ecology. In the US in the 1960s, Edward O. Wilson and Robert MacArthur discovered key factors determining species diversity, immigration, and extinction on islands. James Brown later did similar work on animal populations in isolated patches of forest ridge in California. Such work has showed how to identify species most at risk of extinction due to isolation.

Stability and resilience

One major contribution to the understanding of ecosystem dynamics was the concept of the evolutionarily stable state. In the 1970s, British biologist John Maynard Smith used the term evolutionarily stable strategy (ESS) to describe the best behavioral strategy for an animal competing with others living in its vicinity. This strategy depends on how the other animals behave and is determined by the animal’s genetic success—if it makes the wrong